

Software Requirements Specification (SRS)

Project: TechConnect

Version: 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete set of functional, non-functional, user, system, and data requirements for **TechConnect** – a digital marketplace that connects skilled individuals with paying clients in a structured, secure environment. It is intended for the development team, project stakeholders, and quality-assurance testers to guide design, implementation, and validation.

1.2 Scope

The system will:

- Allow users to register with distinct roles (Skilled Individual / Client).
- Enable skilled users to create profiles, list skills (predefined or custom), and build a public reputation.
- Allow clients to post jobs with budgets, milestones, required skills, and deadlines.
- Provide intelligent matching between jobs and qualified freelancers.
- Facilitate a milestone-based workflow with secure, escrow-like payments via **Nkwa Pay**.
- Incorporate a 1-5 star rating and review system for both parties after job completion.
- Include an admin dashboard for user management, dispute resolution, and transaction monitoring.
- Deduct a standard 10% platform commission from each completed transaction before releasing earnings to the freelancer.
- Operate initially in Cameroon (FCFA currency) with full web-based access, later expandable.

1.3 Definitions & Acronyms

Term	Definition
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Term	Definition
Skilled Individual (Freelancer)	A person who offers a service (coding, design, writing, etc.) on the platform.
Client	A person or entity that posts a job and pays for the service.
Nkwa Pay	The integrated payment gateway that handles transactions and escrow.
Escrow	A financial arrangement where funds are held by the platform until work is approved.
Milestone	A discrete deliverable within a job, used to stage work and payments.
Commission	A 10% fee deducted from a freelancer's earnings before payout.
Rating	A 1-5 star evaluation given by both parties after job closure.
Admin	Platform manager with privileges to resolve disputes and oversee transactions.

2. Overall Description

2.1 Product Perspective

TechConnect is a **web-based marketplace** built on a cloud-hosted low-code/no-code infrastructure (Bubble) for the initial minimum viable product (MVP). It interacts with the Nkwa Pay API for all financial operations. The system is accessible through modern browsers on desktops, tablets, and mobile devices. It replaces informal, insecure, and trust-less ways of finding freelance work in African markets.

2.2 Product Functions (Summary)

- User registration & role selection
- Skill & profile management
- Job posting with milestones and budget
- Smart matching & manual search
- Hiring flow (offer, accept, decline)
- Milestone-by-milestone submission, review, and approval
- Escrow payment via Nkwa Pay (full upfront or milestone-based)
- Automated commission deduction
- Dual-party rating & feedback
- Admin dispute intervention
- Transaction logging & audit

2.3 User Characteristics

User Group	Description	Key Needs
Skilled Individuals	Coders, designers, video editors, writers, virtual assistants, etc. Often self-taught, with no formal credit history.	Secure payment, access to legitimate clients, reputation building.
Clients	Small businesses, startups, student project owners, local shops, NGOs.	Reliable workers, transparent progress, pay-only-when-satisfied guarantee.
Admin	Platform manager.	Efficient moderation, dispute resolution, oversight of transactions.

All users are assumed to have basic digital literacy, an email address or phone number, and an internet connection.

2.4 Assumptions & Constraints

- Users possess a valid email or mobile number and stable internet access.
 - The Nkwa Pay API is available, reliable, and supports required escrow-like flows.
 - Clients are willing to deposit funds before work begins (full or per milestone).
 - All monetary values are in **Central African CFA franc (FCFA)** during initial rollout.
 - The platform must comply with applicable Cameroonian data protection and e-commerce regulations.
 - The development team uses an Agile (Scrum) methodology with iterative releases.
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3. User Requirements (Non-technical)

These statements describe what users expect from the platform from a non-technical perspective:

- I want to create an account easily using my email or phone number.
 - As a skilled person, I want to showcase my abilities with a profile, portfolio, and skill list.
 - As a client, I want to post a job describing exactly what I need, my budget, and deadlines.
 - I want to see a list of jobs that match my skills or, if I'm a client, browse available freelancers.
 - I want a safe payment method where the money is only released after I confirm satisfactory work.
 - I want to be rated for my work so I can attract more clients in the future.
 - As a client, I want to see a freelancer's past ratings and feedback before I hire.
 - I want to break large projects into small steps so I can pay gradually and reduce risk.
 - If something goes wrong, I want a fair administrator to help resolve the dispute.
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4. Functional Requirements

All functional requirements are expressed using the phrase **"The system shall..."**. They are grouped by module.

4.1 User Registration & Authentication

1. The system shall allow a user to register using an email address and password.
2. The system shall verify the email or phone number via a one-time code (OTP) before account activation.
3. The system shall require the user to select a role: **Skilled Individual** or **Client** during registration.
4. The system shall store passwords using a strong one-way hashing algorithm (e.g., bcrypt).
5. The system shall provide password reset functionality via the registered email.

4.2 Profile & Skill Management

6. The system shall allow a Skilled Individual to create a public profile containing a short biography, profile picture, and a portfolio of past work (images, links, videos).
7. The system shall present a predefined list of skills (e.g., "Logo Design", "Python", "Social Media Management") and allow the user to add custom skills.
8. The system shall display the user's skills on their public profile.
9. The system shall allow the Skilled Individual to update their profile and skills at any time.

4.3 Job Posting (Opportunities)

10. The system shall allow a Client to post a job with the following mandatory fields: title, description, budget (in FCFA), deadline, required skills, and a milestone structure (if milestone payments are chosen).
11. The system shall validate that the budget is a positive number and the deadline is in the future.
12. The system shall allow the Client to attach reference files or links to the job posting.
13. The system shall store the job with an initial status of **"Created"**.

4.4 Matching & Discovery

14. The system shall automatically recommend jobs to Skilled Individuals based on matching skills.
15. The system shall provide a search interface where freelancers can filter jobs by skill, budget range, or keyword.
16. The system shall display a freelancer's profile and rating to clients who are searching for talent.

4.5 Hiring & Acceptance

17. The system shall allow a Client to send a formal job offer to one or more Skilled Individuals.
18. The system shall notify the selected freelancer of the offer.
19. The system shall allow the freelancer to **accept** or **decline** the offer within 48 hours.
20. Upon acceptance, the job status shall change to **"Accepted"** and the parties are contractually bound through the platform.

4.6 Milestone-Based Workflow

21. The system shall require each job to be divided into one or more milestones (at least one milestone if the job is not a single-payment task).
22. The system shall allow the freelancer to submit work deliverables for a specific milestone (including text description and file uploads).
23. The system shall notify the client when a milestone is submitted.
24. The system shall allow the client to either **"Approve"** the milestone or **"Request Changes"** with a mandatory comment.
25. If changes are requested, the freelancer shall resubmit; the system shall limit the number of revision rounds to three per milestone to prevent endless loops.
26. After the final milestone is approved, the system shall change the job status to **"Completed"**.

4.7 Payment Flow (Nkwa Pay Integration)

The entire payment flow ensures an escrow-like model.

27. The system shall allow the client to choose either **"Full Upfront Payment"** for small jobs or **"Milestone-Based Payment"** for larger projects, with the total budget split among milestones.
28. **Payment Initiation:** Before any work begins, the client shall be redirected to Nkwa Pay to deposit the required amount (full or milestone total) into the platform's escrow account.
29. The system shall not activate the job (status: **"Active"**) until a successful payment confirmation is received from Nkwa Pay.
30. On successful payment, the system shall:
 - Verify the transaction via the Nkwa Pay API success callback or status check.
 - Store the unique transaction ID, amount, and timestamp.
 - Update the payment record status to **"Confirmed"**.
 - Notify the freelancer to commence work.
31. **On payment failure**, the system shall:
 - Keep the job status as **"Awaiting Payment"**.
 - Notify the client of the failure with instructions to retry.
 - Log the failed attempt for auditing.
32. **Release of Funds:** After a milestone is approved by the client, or the entire job is marked **"Completed"**:

- The system shall release the corresponding funds to the freelancer via Nkwa Pay disbursement.
- The system shall automatically deduct a **10% platform commission** from the released amount before the freelancer receives it (e.g., for a 20,000 FCFA milestone, the freelancer receives 18,000 FCFA).

33. **Dispute Intervention:** If a milestone or job completion is disputed, the system shall freeze the escrowed funds and notify the Admin. Only the Admin can release funds after mediation.

4.8 Rating & Feedback

34. The system shall prompt both the Client and the Skilled Individual to rate each other on a 1-5 star scale and leave an optional written review within 7 days of job completion.

35. The system shall compute an aggregate rating score for each user and display it on their profile.

36. The system shall use rating data to influence search ranking and trust indicators (e.g., “Top Rated” badge for users with an average ≥ 4.5 over at least 10 jobs).

4.9 Admin Functions

37. The system shall provide an Admin dashboard to view all users, jobs, payments, and disputes.

38. The system shall allow the Admin to suspend or deactivate user accounts after investigation.

39. The system shall allow the Admin to view full payment logs and manually override a fund release in the event of a deadlocked dispute.

40. The system shall notify the involved parties of any administrative action.

4.10 Data Storage Requirements

The following core entities shall persist in the database, with the relationships described.

Entity	Key Attributes
User	userID, name, email, hashedPassword, role (skilled/client/admin), rating, bio, profileImageURL
Skill	skillID, skillName, category, userID (foreign key)
Job	jobID, title, description, budget, status, clientID, skilledUserID, deadline, creationDate
Milestone	milestoneID, jobID (FK), description, amount, status, submissionFileURL, dueDate
Payment	transactionID (from Nkwa Pay), jobID (FK), milestoneID (FK, nullable), amount, status (pending/confirmed/released/failed), commission, date
Rating	ratingID, jobID (FK), fromUserID, toUserID, score (1-5), comment, date

Relationships:

- One User → many Skills
 - One Job → one Client + one Skilled Individual
 - One Job → many Milestones
 - One Job → one or more Payments (each milestone can have a payment)
 - Each completed Job generates exactly two Rating records.
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5. Non-Functional Requirements

5.1 Security

- All passwords shall be stored using a strong salted-hash algorithm (bcrypt, minimum work factor 10).
- Payment and personal data shall be encrypted in transit (TLS 1.2+) and at rest.
- The system shall implement role-based access control (Skilled, Client, Admin) for all endpoints.

5.2 Reliability & Availability

- The system shall be available 24/7 with a target uptime of 99.5%.
- Payment statuses must be accurate and consistent; all failed transactions shall be logged with the reason code from Nkwa Pay.

5.3 Performance

- All web pages shall load within 3 seconds under normal network conditions (3G mobile connection).
- The system shall support at least 500 concurrent users without degradation.

5.4 Usability

- The interface shall be responsive and usable on screens from 320px width (mobile) to 1920px (desktop).
- All forms shall provide clear validation errors in simple English (or French, if required).
- The platform shall use intuitive icons and a consistent colour scheme to guide first-time users.

5.5 Maintainability

- The system logic shall be modular, with clear separation between UI, business logic, and integration layers.

- All Nkwa Pay API interactions shall be isolated in a dedicated service module to ease future changes of payment providers.

5.6 Regulatory

- The system shall comply with Cameroon's Law No. 2010/012 on data protection where applicable, and obtain user consent for data processing.
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6. System Requirements / Software Requirements

6.1 Development Tools

- **Application Builder:** Bubble (low-code/no-code platform) for rapid prototyping and MVP deployment.
- **Browser-based testing:** Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions).
- **Version Control:** Git repository for any custom code components, integrated with Bubble if required.
- **Collaboration:** Project management via Trello or Jira; team communication via Slack.

6.2 Hardware Requirements (User Side)

- A laptop, desktop computer, tablet, or smartphone.
- Minimum 4 GB RAM recommended.
- Stable internet connection (mobile data or broadband).

6.3 Software Environment

- **Hosting:** Cloud-based (Bubble's Amazon Web Services infrastructure); no local installation required.
 - **Client-side:** Any modern web browser with JavaScript enabled.
 - **No app store download:** The platform is accessible entirely via a secure URL (HTTPS).
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7. System Design Overview

While detailed design is out of scope for an SRS, a high-level architecture is outlined to clarify system boundaries and interactions.

Components:

- **User Device:** Smartphone, tablet, or PC running a web browser.
- **Frontend (UI Layer):** Bubble-built responsive interface with buttons, forms, job cards, and dashboards. Communicates with the backend via RESTful APIs.
- **Backend (Application Logic):** Bubble's workflow engine manages user sessions, job matching algorithms, milestone state machines, and rating calculations. Custom JavaScript/API Connector handles Nkwa Pay.
- **Nkwa Pay API (External):** Handles payment initialization, transaction confirmation, and disbursement. The platform never stores raw payment credentials.
- **Database:** Bubble's integrated database (backed by PostgreSQL logic) stores all user, job, payment, and rating records in a structured, relational model.

Data Flow (Simplified):

1. Client initiates payment → frontend calls backend → backend requests Nkwa Pay checkout.
 2. Nkwa Pay callback → backend verifies → updates payment record → notifies freelancer.
 3. Milestone approval → backend triggers Nkwa Pay disbursement (minus commission) → freelancer receives funds.
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8. Development Methodology

The development team will follow the **Agile (Scrum)** model:

- The project is divided into 2-week sprints.
 - An MVP will be delivered first, containing core features: registration, job posting, milestone-based hiring, and basic Nkwa Pay integration.
 - Early feedback from a pilot group of Success's university network will be collected after each sprint.
 - Iterative improvements will add the rating system, admin dashboard, and advanced matching.
 - Agile fits the team size of 5, allows rapid adaptation to changing requirements, and reduces risk through continuous testing.
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9. Future Improvements

Features planned for future releases after the successful launch of the MVP:

- **Real-time chat** between clients and freelancers for clarifications.
- **Mobile-first progressive web app (PWA)** and later native Android/iOS apps.
- **Multi-currency and multi-payment gateway support**, including MTN Mobile Money and Orange Money.
- **Identity verification** (government ID upload with selfie) for a "Verified" badge.

- **Skill assessment tests** to earn “Certified” badges for high-demand skills.
 - **Tiered commission levels** to reward high-volume or long-term users.
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10. Additions Inspired by Market Research

To ensure TechConnect is competitive and delivers an excellent user experience in African markets, the following features were added beyond the original concept, inspired by successful regional platforms like **Terawork** (Nigeria) and **KaziWorks** (Kenya):

Addition	Rationale
Portfolio section	Freelancers need more than a list of skills; they must showcase real work. Terawork allows portfolios, which dramatically increases hire rates.
“Top Rated” badge (auto-awarded)	Gamification and visible trust signals. Similar to Upwork’s “Top Rated” but tailored for small markets where a few stars can decide a hire.
Dispute freeze & admin mediation	Escrow alone isn’t enough; a clear admin-mediated dispute path is essential in low-trust environments. Terawork offers a “Resolution Center” that reduces litigation.
Client verification optional flag	While not mandatory, clients can request verified freelancers only (future feature), similar to Terawork’s verified talent pool.
Responsive design as a non-functional requirement	Most users in Cameroonian cities browse via mobile. Research on Jobberman and Terawork shows >70% mobile traffic; thus, mobile-first UX is critical.

These additions enhance trust, safety, and user experience without increasing implementation complexity unreasonably, because they leverage the core infrastructure already planned.

Prepared for: TechConnect Development Team, LMUI Introduction to Software Engineering lecturer Mr. Meh Armstrong

Based on: Original analysis document, stakeholder discussions, and market research on trusted African freelance platforms.